

# Problem Statement



# Problem Statement - Background



## **Blood Warriors**

- Blood Warriors is a foundation connecting voluntary blood donors with Thalassemia patients across India. With over 1,00,000 patients requiring up to 500 to 700 transfusions in a lifetime, the demand for reliable, recurring, and compatible blood is constant. Their Blood Bridge initiative and donor network are at the heart of making this happen.

## **The Problem**

- Today, matching donors to patients, coordinating requests, and keeping the ecosystem engaged relies heavily on manual effort. As Blood Warriors scales across cities and partner organizations, this becomes harder to sustain.

## **AI-Enabled Care Coordination and Access**

- How might we make it easier for different people involved in care to stay connected and informed at the right time?
- What could more intuitive, adaptive interactions between individuals and systems look like across languages and contexts?
- How might systems remember and respond appropriately over repeated interactions?

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## **Smart Matching and Resource Allocation**

- How can we better identify and prioritize the right people or resources when needs arise?  
What signals or patterns could be used to anticipate availability or willingness to participate?

## **Engagement and Continuity**

- How might we encourage continued participation and build long term engagement among contributors?
- What kinds of feedback loops or interactions could make the system feel more responsive and human?

## **Operating at Scale**

- What would it take for such a system to function effectively across a large, diverse population?
- How might it adapt to variations in language, access, and infrastructure without losing efficiency?
- What considerations come into play when expanding to support many users and regions simultaneously?

# Problem Statement



Build an autonomous, AI-powered blood support network that can coordinate requests, engage the right participants, and manage interactions in real time with minimal manual effort.

## **What the System Should Be Able to Do**

- Handle multiple workflows through a unified intelligent AI layer
- Automate outreach, follow-ups, and escalations for donors
- Track and interpret donor responses to guide next steps
- React to real-time events and updates
- Enable conversational interactions with memory
- Systems must self manage improvement steps and protocols via failure learning
- Provide admins with dashboards and insights
- Ensure consent-aware, responsible data usage and compliant systems

# Infrastructure



# Infrastructure - AWS

The following **AWS Services** are authorized for use alongside **GitHub** integrations:

## **Compute & Web**

- S3 + CloudFront / Amplify
- AWS Cognito
- AWS App Runner
- Lambda + API Gateway
- Fargate / Amazon EKS

## **Events & Streams**

- Amazon SQS / SNS / SES
- Amazon EventBridge
- IAM / CloudWatch
- Kinesis Data Streams
- AWS Glue

## **Databases**

- Amazon DynamoDB
- RDS (Postgres / Aurora)
- Amazon Redshift
- Amazon Athena
- OpenSearch Service

## **Security & Ops**

- Secrets Manager / Parameter Store
- WAF / AWS Shield
- AWS X-Ray
- Managed Airflow (MWAA)

## **AI & Machine Learning**

- Amazon Bedrock
- Amazon SageMaker
- Amazon Lex
- SageMaker Feature Store

## **DevOps**

- AWS CodeBuild
- AWS CodePipeline
- AWS CodeCommit
- GitHub Actions

We will send a soft notification when usage goes beyond \$30, followed by another notification if it exceeds \$40. Participants will be asked to downgrade services (e.g. Save work on GitHub) upon exceeding \$40.

# Recommended Tech Stack

1

## Full Stack (User Interaction Layer)

### Recommended:

Front-end:  
React.js (web UI)  
Back-end:  
Python (Fast API / Flask), REST APIs + Async processing

2

## Data Engineering (Pipelines and Storage)

### Recommended:

Database: AWS Databases (RDS, S3, Aurora, DynamoDB)  
ETL pipelines services via AWS(Glue, Kinesis)

3

## Data Science & AI (Intelligence Layer)

### Recommended:

ML: Matching / Ranking models, NLP, Predictions Layers(SageMaker)  
AI: Conversational AI(BedRock)

4

## Automation Systems (Orchestration Layer)

**Recommended:** AWS Step function, AWS Lambda, API Gateway

5

## Deployment and Monitoring

### Recommended:

AWS EC2, CloudWatch  
**Optional:** ECS + AWS CodePipeline



# Hackathon Flow



# AI For Good 2.0 : Overall SCHEDULE

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Two-day schedule at a glance

## DAY 1

### START

**11:00 AM**

Kick-off

### CHECKPOINT

**04:00 PM**

Checkpoint 1

### Ice Breaker

**07:00 PM**

Ice Breaker Session 1

## DAY 2

### CHECKPOINT

**09:00 AM**

Checkpoint 2

### DEADLINE

**11:00 AM**

Submission

### JUDGEMENT

**01:30 PM**

Jury Session

# Day 1 · Detailed Schedule

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|                           |                                         |                         |                             |
|---------------------------|-----------------------------------------|-------------------------|-----------------------------|
| <b>8:00 – 10:00 AM</b>    | Registrations & Check-in                | <b>3:30 – 4:00 PM</b>   | Hi-Tea & Snacks             |
| <b>8:30 – 9:30 AM</b>     | Breakfast                               | <b>4:00 PM Onwards</b>  | 1st Checkpoint & Mentorship |
| <b>10:00 – 11:00 AM</b>   | Opening Ceremony, Keynote, Ground rules | <b>4:00 – 7:00 PM</b>   | Hack Hours                  |
| <b>11:00 AM</b>           | Hackathon Kick-off                      | <b>7:00 – 7:30 PM</b>   | Ice Breaker Session 1       |
| <b>11:00 AM – 1:00 PM</b> | Hack Hours                              | <b>7:30 – 9:00 PM</b>   | Hack Hours                  |
| <b>1:00 PM</b>            | Lunch                                   | <b>9:00 PM</b>          | Dinner                      |
| <b>2:00 – 3:30 PM</b>     | Hack Hours                              | <b>10:00 – 11:59 PM</b> | Hack Hours                  |

# Day 2 · Detailed Schedule

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|                            |                                 |                        |                                   |
|----------------------------|---------------------------------|------------------------|-----------------------------------|
| <b>12:00 – 12:30 AM</b>    | Icebreaker Session 2            | <b>12:00 PM</b>        | Top 15 Team Announcements         |
| <b>12:30 – 8:00 AM</b>     | Hack Hours                      | <b>12:00 – 1:00 PM</b> | Lunch                             |
| <b>8:00 – 9:00 AM</b>      | Breakfast                       | <b>1:00 – 4:30 PM</b>  | Jury Sessions                     |
| <b>9:00 AM Onwards</b>     | 2nd Checkpoint & Mentorship     | <b>4:30 – 6:00 PM</b>  | Closing Keynote Address           |
| <b>9:00 – 11:00 AM</b>     | Hack Hours                      | <b>4:30 – 6:00 PM</b>  | Announcement of Winners & Results |
| <b>11:00 AM</b>            | Hackathon Ends with Submissions | <b>4:30 – 6:00 PM</b>  | Vote of Thanks                    |
| <b>11:00 AM – 12:00 PM</b> | Break                           |                        |                                   |